CS 355 - Topic 9: Routing

Dates: March 25, 2026

March 25:

Routing Protocols

- Decide which router / network to forward the message to next
- EGP - External Gateway Protocol
    - EGPs is used to forward messages between 2 or more autonomous systems
    - Autonomous Systems: collection of networks owned and managed by a single operator
        - Autonomous Systems are mainly ISPs but also can be large companies such as Microsoft
    - Protocol: BGP Border Gateway Protocol
- IGP - Internal Gateway Protocol
    - IGPs are used to forward messages inside a single autonomous system
    - Protocols: OSPF, IS-IS, RIP

EGP Relationships

- Peering Relationship

    - Two ISPs (e.g., ISP A and ISP B) agree to exchange traffic between their customers
    - Example: C1, C2 connect through ISP A ⇄ ISP B to reach C3, C4

- Transit Relationship

    - A company pays an ISP to carry its traffic
    - Example: Company X uses ISP B as a transit provider → leads to ISP B using Company X to help improve infrastructure

BGP — Best Path Selection Algorithm

- Proceed until the tie is broken
- Autonomous systems have a unified routing policy
- IGP prioritizes performance, which boils down to the shortest path generated by Dijkstra's Algorithm (solves the single-source shortest path problem)

OSPF — Open Shortest Path First

- Each router will periodically, and through some events, broadcast LSAs (Link State Advertisements) to its neighbors
- LSAs contain router + network connectivity information
- Each LSA received updates a LSDB (Link State Database) until the LSA no longer provides new information (at which point broadcasting stops)
- The LSDB converges to the full graph topology for time t